15. The Synthetic Control Method

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Synthetic Control Method* (Prof. Sanghack Lee, GSDS SNU; drawing on Alberto Abadie and Matthew Blackwell)

The central question is how to estimate the effect of an intervention on a *single* treated unit (a state or country) when no single untreated unit is a credible comparison, by building a weighted combination of untreated units that tracks the treated unit before the intervention.

These notes consolidate the slides, state the estimator and its model-based justifications, explain convexity, sparsity and placebo inference, derive the bias from imperfect pre-treatment fit, and present augmented SCM and the choice of predictor weights. Physical PDF page numbers are listed in Section C. References: Abadie and Gardeazabal (2003, AER); Abadie, Diamond and Hainmueller (2010, JASA); Abadie (2021, JEL); Ben-Michael, Feller and Rothstein (2021, JASA).

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1 From DiD to synthetic controls

Purpose. Motivate a data-driven comparison unit when there is only one treated unit.

DiD works well with two groups and parallel trends. But with a single treated unit (a country adopting a policy), which control should be used? Any single control may not be comparable. The **synthetic control method** (SCM) constructs an optimal weighted combination of control units to approximate the counterfactual.

Abadie and Gardeazabal (2003) used a DiD-like approach for "quantitative case studies": an intervention in one country/state at one point in time; one treated group and many controls; compare time series of the treated unit with a control. Since there are many possible control choices, use a *convex combination* of the controls, a synthetic control, with weights minimizing pre-treatment differences between the treated unit and its synthetic counterpart.

1.1 Case study: California Proposition 99

In 1988 California passed Proposition 99, a \$0.25 per pack cigarette tax increase (the largest state tax increase at that time). Goal: the effect on per-capita cigarette sales. No single state is a good comparison. SCM builds a "synthetic California" from 38 control states; key donors are Utah (33.4%), Nevada (23.4%), Montana (19.9%) and Colorado (16.4%). Weights are chosen to match pre-1988 cigarette sales, income, beer consumption, retail price and the share of population aged 15–24. Result: by 2000 California's per-capita sales were about 26 packs lower than synthetic California's (Abadie, Diamond and Hainmueller 2010). In the figure, actual and synthetic California track each other closely from 1970 to 1988 and diverge afterwards. Without synthetic controls, California compared with the average of the other states differs already before 1988; the synthetic control reproduces California's pre-1988 smoking and predictor values, which, like matching, forces the researcher to demonstrate the affinity between the treated unit and its counterpart and guards against "extreme counterfactuals" far outside the convex hull of the data.

2 The estimator

Purpose. Define the setting, estimand and weights.

2.1 Setup

Time period	1	2	$\cdots$	T_0	T_0+1	$\cdots$	T
Treated unit ($i=1$)	0	0	0	0	1	1	1
Control group ($i=2,\ldots,J+1$)	0	0	0	0	0	0	0

All units are untreated for T_0 periods; unit 1 is treated from T_0+1 to T. Potential outcomes $Y_{it}(x)$, $x \in \{0,1\}$; observed $Y_{it} = Y_{it}(X_{it})$. Estimand: the effect on the treated unit over time,

$$\tau_{1t} = Y_{1t}(1) - Y_{1t}(0), \qquad t > T_0.$$

2.2 Missing counterfactuals and synthetic weights

By consistency, for $t > T_0$, $\tau_{1t} = Y_{1t} - Y_{1t}(0)$, so the missing $Y_{1t}(0)$ must be imputed. Choose weights $(w_2, \ldots, w_{J+1})$ with

$$w_j \geq 0, \qquad \sum_j w_j = 1 \quad \text{(convex)},$$

minimizing, over pre-treatment periods $t \leq T_0$,

$$\left\| Y_{1t} - \sum_{j=2}^{J+1} w_j Y_{jt} \right\| \quad \text{and} \quad \left\| Z_1 - \sum_{j=2}^{J+1} w_j Z_j \right\|,$$

where Z_i are pre-treatment predictors. The objectives are combined via a weighted norm with predictor-importance weights V (Appendix B); a penalty on the dispersion of the weights can be added. The hope is that, for $t > T_0$, $\sum_{j=2}^{J+1} w_j Y_{jt} \approx Y_{1t}(0)$, giving $\hat{\tau}_{1t} = Y_{1t} - \sum_j w_j Y_{jt}$.

2.3 Model-based justifications

Model 1: interactive factor model.

$$Y_{it}(0) = Z_i^\top \theta_t + \alpha_i + \delta_t + \lambda_t \mu_i + \varepsilon_{it},$$

with θ_t time-varying coefficients on covariates, α_i a unit fixed effect, δ_t a common time shock, λ_t a $1 \times F$ vector of common factors (unobserved macro shocks) and μ_i an $F \times 1$ vector of factor loadings (each unit's sensitivity to those shocks). The term $\lambda_t \mu_i$ allows time-varying confounding in a structured way; DiD is the special case without it.

Model 2: autoregressive model (no fixed effects):

$$Y_{i,t+1}(0) = \alpha_t Y_{it}(0) + \beta_{t+1}^\top W_{i,t+1} + u_{i,t+1}, \qquad W_{i,t+1} = \gamma_t Y_{it}(0) + \Pi_t W_{it} + v_{i,t+1},$$

with time-varying covariates W_{it}, AR coefficient α_t (the dynamic source replacing the factor structure), coefficients β_{t+1}, feedback γ_t from outcome to covariates, and autoregressive matrix Π_t.

Properties. Suppose perfect balancing weights exist: $\sum_j w_j Y_{jt} = Y_{1t}$ for $t \leq T_0$ and $\sum_j w_j Z_j = Z_1$. Let $\hat{Y}_{1t}(0) = \sum_j w_j Y_{jt}$ for $t > T_0$.

- Under Model 1, $\hat{Y}_{1t}(0) \to Y_{1t}(0)$ as $T_0 \to \infty$: consistent, improving as the pre-intervention period grows (matching many pre-period outcomes forces $\sum_j w_j \mu_j \approx \mu_1$).
- Under Model 2, $\mathbb{E}[\hat{Y}_{1t}(0)] = \mathbb{E}[Y_{1t}(0)]$ (given perfect fit): unbiased even with a single pre-treatment period.

2.4 Convexity: no extrapolation

SCM requires $w_j \geq 0$ and $\sum_j w_j = 1$, so the synthetic control is a convex combination of donors. Example: California's pre-treatment smoking is 120 packs per capita, while all donors lie between 60 and 110; convex weights can reach at most 110, so the imperfect fit is visible and transparent. OLS, in contrast, may assign negative weights (e.g. $w_j = -0.3$); the weights still sum to one, but the synthetic outcome can lie far outside the observed range, and this extrapolation is hidden. SCM trades flexibility for transparency: one can always check whether the treated unit lies inside the convex hull of the donors.

Sparsity: geometric intuition. SC weights are sparse: most $w_j = 0$. Geometrically, the treated unit is approximated by a point on the boundary of the convex hull of donors, so only a few "vertex" donors get positive weight, which aids interpretation. All weight decisions are made without post-treatment data, protecting against specification searching.

2.5 Practical concerns

- To reduce interpolation bias and overfitting, restrict the donor pool to units similar to the treated unit.
- Donors must not be affected by the treatment (no interference or spillovers).
- "Although we define our synthetic controls as convex combinations of unexposed units, negative weights or weights larger than one can be used at the cost of allowing extrapolation." (Relaxing the no-extrapolation property.)
- Convex weights limit flexibility: the treated unit's pre-treatment outcomes may not be matched perfectly, especially if it lies outside the convex hull; the trade-off is deliberate, for interpretability and robustness.
- With $N = 1$ treated unit, inference uses placebo tests.

3 Inference and bias

Purpose. Assess significance by placebo permutation, and understand bias from imperfect fit.

3.1 Placebo tests

With one treated unit, sampling-based inference is unavailable. Iteratively apply SCM to each control state as if it were treated: for each of the 38 control states, build a synthetic control and estimate a "placebo effect". If SCM works well, most placebo effects are near zero. This is a permutation/randomization-inference argument: under the null of no effect, California's gap should look like a typical draw from the distribution of placebo gaps; an unusually large gap is evidence of a real effect. The "p-value" is the fraction of placebo gaps at least as large as California's. In Abadie et al. (2010) only 1 of 34 states had a larger gap, $p \approx 0.03$.

In the placebo plot, gray lines show gaps $Y_{jt} - \hat{Y}_{jt}^{\mathrm{SC}}$ for control states and a red line California's gap: most placebo gaps are small and centered at zero, while California's diverges sharply after 1988. Refinement: exclude states with poor pre-treatment fit (high pre-treatment RMSPE), whose placebo gaps are noisy and uninformative.

3.2 Why imperfect pre-treatment fit causes bias

Under Model 1, if $\sum_j w_j Y_{jt} = Y_{1t}$ for all $t \le T_0$, the factor loadings are approximately matched, $\sum_j w_j \mu_j \approx \mu_1$. If the match is imperfect for some $t \le T_0$, then $\sum_j w_j \mu_j \neq \mu_1$, and post-treatment the common factors create a systematic wedge:

$$\sum_j w_j Y_{jt}(0) - Y_{1t}(0) \approx \lambda_t \Big(\sum_j w_j \mu_j - \mu_1 \Big) \neq 0,$$

the other terms ($Z_i^\top \theta_t$, α_i, δ_t) cancelling by predictor balancing and ε_{it} averaging out. Intuition: the synthetic control does not capture unit 1's response pattern to macro shocks, so predictions drift after treatment. This is especially problematic when the treated unit lies outside the convex hull of donors.

3.3 Bias correction: augmented SCM

Use an outcome model to correct residual mismatch (Ben-Michael, Feller and Rothstein 2021). Model $Y_{it}(0) = m_{it} + \varepsilon_{it}$; fit the regression on control units only ($j = 2, \ldots, J+1$) and let $\hat{m}_{it}$ denote predictions, including for $i = 1$. The augmented estimator is

$$\hat{Y}_{1t}^{\text{aug}}(0) = \underbrace{\sum_{j=2}^{J+1} w_j Y_{jt}}_{\text{SC estimate}} + \underbrace{\Big(\hat{m}_{1t} - \sum_{j=2}^{J+1} w_j \hat{m}_{jt}\Big)}_{\text{bias correction}} = \underbrace{\hat{m}_{1t}}_{\text{outcome model}} + \underbrace{\sum_{j=2}^{J+1} w_j (Y_{jt} - \hat{m}_{jt})}_{\text{SC-weighted residuals}}.$$

Covariates are easy to add. It is doubly robust: consistent if either the SC weights or the outcome model is approximately correct (analogous to AIPW and doubly robust DiD).

4 Discussion

Purpose. Weigh strengths and limitations and position SCM among alternatives.

Strengths	Limitations
Transparency: weights are explicit; one can inspect which donors contribute and how much.	Needs a good donor pool: similar units with no spillovers.
Data-driven: no need to pick a single comparison unit by hand.	Designed for a single treated unit; extensions needed for several.
No parallel-trends assumption (uses a factor model instead).	Sensitive to donor-pool and predictor choice.
Pre-treatment fit is a built-in diagnostic of credibility.	Formal inference is limited (permutation-based, not sampling-based); convexity can prevent good fit when the treated unit is an outlier.

(Abadie 2021.)

When to use SCM: one or very few treated units with multiple controls; a rich pre-treatment period; parallel trends implausible but a factor structure reasonable. Consider DiD with multiple treated and control units and plausible parallel trends; consider synthetic DiD to combine both, especially in larger panels. Open research areas: formal inference beyond permutation tests (conformal inference), sensitivity analysis for SCM specifications, experimental design with synthetic controls, external validity.

Extensions. Augmented SCM (bias correction, double robustness); synthetic difference-in-differences (Arkhangelsky et al. 2021: unit weights as in SCM, time weights and unit fixed effects as in DiD, valid large-panel inference); penalized SCM (Abadie and L'Hour 2021: relax strict convexity with L_1/L_2 penalties, allowing regularized partial extrapolation); conformal inference for SCM (Chernozhukov, Wüthrich and Zhu 2021: distribution-free prediction intervals via permutation of residuals); matrix completion (Athey et al. 2021: the counterfactual as a missing-data problem with a low-rank approximation).

DiD versus synthetic control.

	Difference-in-differences	Synthetic control
Setup	two groups, two (or more) periods	one treated unit, many donors
Key assumption	parallel trends	pre-treatment fit (factor model)
Estimand	ATT (group level)	τ_{1t} (unit level)
Strength	simple, transparent	handles one treated unit well
Weakness	sensitive to nonlinear transforms	requires a good donor pool
Extensions	staggered DiD, nonlinear DiD	augmented SCM, matrix completion

A More treated units

Two estimation approaches generalize to any number of treated units. *Interactive fixed effects:* with $Y_{it}(0) = Z_i^\top \theta_t + \alpha_i + \delta_t + \lambda_t \mu_i + \varepsilon_{it}$ (Model 1), estimate the IFE directly instead of weighting, by iterating:

(1) treat the IFE terms as fixed and fit the parametric part on untreated observations to get $\hat{\theta}$;

(2) treat the covariate coefficients as fixed and estimate the IFE terms by factor analysis (principal components of residuals);

(3) repeat until convergence. *Matrix completion* (Athey et al. 2021): treat the matrix of control potential outcomes $\mathbf{Y}(0)$ as partially missing and estimate a low-rank matrix $\mathbf{L}$ approximating its observed entries, with regularization (as in recommendation systems).

B Predictor importance weights V

The unit weights minimize a V-weighted predictor mismatch:

$$w(V) = \underset{w \geq 0,\ \sum_j w_j = 1}{\arg\min} \sum_{m=1}^{k} v_m \Big(X_{1m} - \sum_{j=2}^{J+1} w_j X_{jm} \Big)^2,$$

where $v_m \geq 0$ says how much predictor m (a covariate or a pre-period outcome) matters. Simple choices: equal weights $v_m = 1$; variance scaling $v_m = 1/\operatorname{Var}(X_m)$ (common scale); regression-based $v_m \propto |\hat{\beta}_m|$ from an outcome regression on donors. Abadie's standard, automatic choice is nested optimization: $V^* = \arg\min_V \mathrm{MSPE}_{\mathrm{pre}}(w(V))$, the outer loop choosing V to minimize pre-treatment outcome MSPE and the inner loop solving for $w(V)$.

Caveats. (1) *Outcome-lag collapse* (Kaul, Klößner, Pfeifer and Schieler 2022): including all pre-period outcome lags drives $v_m \to 0$ for the other predictors, so matching collapses to outcomes only and covariates are ignored; use only a few lags (e.g. every five years).

(2) *Sensitivity:* V depends on the predictor list and lag choice, and different specifications give different donor weights. Practice: choose predictors theoretically related to the outcome; check robustness across several V methods (equal, variance, nested) and report whether the effect is stable; report donor weights explicitly so readers see which donors drive the result.

C Source coverage

The source deck has 51 physical pages (26 numbered slides).

Topic	Physical pages
References, motivation, Prop 99 figure and case study	1–12
Setup, weights, figures, models, properties	13–21
Convexity, sparsity, practical concerns	22–28
Placebo inference, bias from imperfect fit, augmented SCM	29–37
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